MAT 017B Quiz 5

August 23, 2018

Name: Key

SID:

1. Find a Taylor polynomial of degree n of the function $f(x) = xe^x$.

$$f'(x) = xe^x + e^x$$

$$f''(x) = xe^x + e^x + e^x$$

$$= xe^x + 2e^x$$

$$f'''(x) = xe^x + e^x + 2e^x$$

$$= xe^x + 3e^x$$

$$\vdots$$

$$f^{(n)}(x) = xe^x + ne^x \Rightarrow f^{(n)}(1) = e^1 + ne^1 = (n+1)e$$

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

$$xe^x = \boxed{\sum_{k=0}^{n} \frac{(k+1)e}{k!}(x-1)^k}$$